

The Gongfu of Xinzuo - Learning Hand Sharpening

Beginner and skill-building guide - editorial review edition

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Chapter 1

Your First Successful Hand Sharpening

The aim of a first sharpening session is simple: make the whole edge cut better from heel to tip.

You do not need a perfect angle, an expensive collection of stones or fast movements. You need one stable stone, one repeatable motion and clear signs that tell you what to do next.

This chapter deliberately teaches a forgiving method. Every normal pass covers the complete edge, from tip to heel. An experienced sharpener may work only on the area that needs correction, but a beginner can easily create sharp and dull zones that way. Removing a tiny amount of extra steel is a reasonable price for a first edge that works evenly.

Use this method on an ordinary double-bevel kitchen knife: a chef's knife, santoku, utility knife, nakiri or paring knife sharpened on both sides. Practise on a sound, inexpensive knife before using your most valuable one.

Do not use this method on:

- a serrated knife;
- a traditional single-bevel yanagiba, deba or usuba;
- a bent or twisted blade;
- a blade with a crack, a deep chip or a badly broken tip;
- a knife whose handle or bolster prevents the heel from reaching the stone.

These cases need a different technique or a repair before normal sharpening.

1.1 What you are actually sharpening

Three parts are easy to confuse:

- The **primary bevel** is the broad surface that makes the blade thinner toward the edge.
- The **microbevel** is the very narrow terminal facet immediately above the cutting edge.
- The **apex** is the line where the two microbevels meet. This is the part that cuts.

In this beginner method, sharpen only the **microbevel**. Do not lay the broad primary bevel on the stone. Grinding the primary bevel changes the thickness and geometry of the blade. That operation is called thinning and belongs to the next chapter.

A dull apex has become rounded, bent or worn. The stone removes steel from the two microbevels until they meet again in a clean line.

When the stone reaches the apex from one side, it often pushes a tiny lip of steel toward the other side. This lip is the **burr**. A light burr along the entire edge tells a beginner that the stone has reached the apex everywhere.

The burr is a progress signal, not the finished edge. It must be removed before the knife is ready.

1.2 What you need

Keep the first setup small:

- one medium water stone, ideally around 800-1200 grit;
- a stone holder, rubber base or damp folded cloth;
- water and a towel;
- a dark permanent marker;
- a pencil for checking stone flatness;
- ordinary printer paper for testing.

A 1000-grit stone is the simplest starting point for a normally dull kitchen knife. It removes steel quickly enough to reach the apex while remaining controllable.

A very coarse stone removes mistakes as quickly as it removes damage. Keep stones below about 500 grit for chips, profile repair and later training. A very fine stone can polish the microbevel without reaching a rounded apex efficiently. It is optional during the first sessions.

If the knife remains unchanged after many correct passes on a medium stone, stop before pressing harder. The knife may need a more effective abrasive, a coarser stone or a repair diagnosis. The next chapter explains those choices.

1.3 Inspect the knife before sharpening

Clean and dry the blade. Look straight along the cutting line and then tilt the apex slowly under a strong light.

Check for:

- bright points or short bright lines along the apex;
- chips or dents;
- a rounded or damaged tip;
- a bend, twist or local hollow in the cutting line;
- corrosion close to the edge;
- a microbevel that is much wider on one side;

- a heel that cannot lie on the stone because the handle or bolster blocks it.

Small reflections normally show dull or rolled areas. Deep chips, cracks, obvious bends and severe profile damage are stop signs for the beginner method.

Make a slow paper cut with the heel, middle and tip separately. Remember or write down where the knife catches. This is your starting map. It lets you compare the same areas after sharpening instead of relying on memory.

1.4 Prepare the stone and your workspace

Read the instructions supplied with the stone. Some water stones need soaking; others need only water on the surface. Soak or wet the stone exactly as its maker directs. Do not assume that every water stone can remain immersed. When an absorbent stone does require soaking, the slowing of air bubbles is a useful sign that its pores have filled with water; the maker's instructions still take priority.

Place the stone lengthwise on a stable support. Push gently on each corner. If it rocks or slides, fix the support before touching the knife to it.

The surface must be reasonably flat. Draw a loose pencil grid across the stone. Rub it against the recommended flattening plate until the lines disappear together. If the centre retains pencil while the ends become clean, the stone is hollow and needs more flattening.

Keep the stone wet during sharpening. Add water when the blade starts to drag, the sound becomes dry or the surface begins to feel sticky. The grey water is normal. It contains loose abrasive and steel removed from the knife; it is not the burr.

Stand or sit so your shoulders remain relaxed and your elbows can move freely. The stone should be high enough that you do not bend deeply over it, but low enough that your shoulders do not rise. Leave enough



Figure 1.1: A stable stone is more important than a fast hand. Secure it before trying to control the angle.

empty space to complete the whole pass without hitting the sink, wall or other objects.

1.5 Find the microbevel with a marker

Look for the very narrow shiny strip immediately above the cutting edge. This is the microbevel. On some knives it is easy to see; on others it may be less than a millimetre wide.

Colour only the last one or two millimetres nearest the edge. Do not colour the broad primary bevel as if it were the surface to sharpen.

Rest the blade lightly on the stone. Raise the spine until the microbevel seems to meet the surface, then make one nearly weightless complete-edge pass. Inspect the ink.

- Ink removed above the microbevel but left at the apex means the spine was too low. The stone did not reach the cutting edge.
- Ink removed only from the extreme apex means the spine was higher than the existing microbevel angle.
- Ink removed across the microbevel and up to the apex means the contact is correct enough to begin.

The exact appearance does not need to be perfect. The important point is that the stone reaches the apex without scratching a wide band across the primary bevel.

Around 15 degrees per side is common for many kitchen knives, but it is not a universal specification. A useful visual approximation is to lift the spine by about one quarter of the blade height. For a 50 mm tall chef's knife, this is about 13 mm. A fixed stack of two coins cannot represent the same angle on knives of different heights and should not be used as a universal gauge.

Use the height estimate only to understand the scale of the movement. The marker tells you more because it reads the microbevel that is already on the knife.

1.6 Learn one complete-edge pass

Hold the handle with your dominant hand. Place two or three fingertips from the other hand on the flat of the blade, safely behind the cutting edge. For the beginner method, keep them near the centre of the blade.

The fingertips rest on the blade and move together with it. Do not slide them from tip to heel while the knife is moving. Local finger pressure will be introduced only after your full-edge movement is reliable.

Start with the tip on the far part of the stone. Because the edge curves upward near the tip, begin with the handle already raised slightly. Draw the knife diagonally across the stone so the contact travels through this sequence:

tip - curved front - middle - heel.

As contact moves from the tip through the curved front, lower the handle gradually. Keep its height almost unchanged as the straighter middle and heel cross the stone. At the end, remove pressure, lift the knife, return to the starting position and repeat.

During the working movement, let the cutting edge travel ahead of the spine, as if trying to slice an impossibly thin layer from the stone. This is called an edge-leading pass. Keep the angle stable and do not let the edge dig into the surface.

Do not grind during the return movement in the beginner method. One deliberate edge-leading movement followed by one unloaded return is easier to control than rapid movement in both directions.

Every normal working pass must include the entire edge. Use the same path each time. Move slowly enough to notice sound, resistance and contact. Speed hides mistakes and adds nothing to the result.

1.6.1 How much pressure to use

Use three simple pressure levels:

1. **Working pressure:** enough to hear and feel the stone cutting, but not enough to bend the blade, move the stone, whiten the fingertips or tighten the shoulders.
2. **Refining pressure:** about half the working pressure once the burr is continuous.
3. **Finishing pressure:** little more than the weight of the knife and the hands guiding it.

If the knife sticks, shakes or changes angle, pressure is already too high. Add water, relax and slow down before trying again.

1.7 The complete beginner method

1.7.1 Step 1: make the starting map

Inspect the clean edge under light and test heel, middle and tip on printer paper. Do not sharpen a cracked, bent, deeply chipped or unusual blade.

1.7.2 Step 2: mark the first microbevel

Colour only the narrow microbevel. Make one light complete-edge pass and adjust the spine until the ink disappears from the microbevel up to the apex from tip to heel.

Do not continue if the stone is scratching the broad primary bevel. Recolour the last one or two millimetres and repeat the contact check.

1.7.3 Step 3: sharpen the first side

Make five slow complete-edge passes. Start with the handle slightly raised at the tip, lower it through the curve and finish at the heel. Keep the same microbevel angle and use controlled working pressure.

Rinse the edge and check for a burr on the side opposite the one you sharpened. Check at the heel, middle, curved front and tip.

If the burr is missing anywhere, make another two or three complete-edge passes. Continue to cover the whole edge, but pay closer attention as the missing area crosses the stone. Inspect again after each small group.

Stop as soon as the first light burr is continuous. Do not make it larger for reassurance.

1.7.4 Step 4: sharpen the second side

Turn the knife over while keeping the handle in your dominant hand. Place the other fingertips safely on the flat and repeat the same complete path from tip to heel.

Work first in a group of five and then in groups of two or three. Stop when the burr has moved to the first side along the complete edge.

The second side may need a different number of passes. Stroke counts create rhythm; the burr tells you whether the apex has been reached.

1.7.5 Step 5: reduce and remove the burr

Keep the complete-edge movement, but reduce both pressure and repetition:

1. three light passes on the first side;
2. three light passes on the second;
3. two lighter passes on the first;
4. two lighter passes on the second;
5. six to ten nearly weightless passes, changing side after every pass.

Rinse and inspect. If a directional burr remains, do not return to heavy pressure. Continue with one nearly weightless pass per side and check again.

This countdown is a starting routine, not proof that the burr has disappeared. The checks below decide when the knife is finished.

1.7.6 Step 6: use a finer stone only after the medium stone succeeds

The first goal is a stable edge from the 1000-grit stone. Do not move to a finer stone to hide an incomplete microbevel.

If the knife already passes the light and paper checks and you have a stone around 3000 grit, clean the knife, hands and holder. Make only alternating complete-edge passes with finishing pressure. Do not raise another noticeable burr.

If you are unsure what the fine stone changed, stop on the medium stone. A clean, well-deburred 1000-grit microbevel is an excellent everyday kitchen edge.

1.7.7 Step 7: clean before testing

Wash and dry the knife. Remove abrasive residue from the handle, blade and work surface. Carbon steel should be dried immediately.

Rinse the stone and let it dry slowly in a ventilated place as its maker recommends. Do not use an oven, radiator, hair dryer or strong direct sun. Do not close a damp stone inside a box.

Test the knife only after it is clean.

1.8 Understand and check the burr safely

The burr is a microscopic lip still attached to the apex. It bends toward the side opposite the one being sharpened. The burred side may feel slightly rough or hooked when approached from the blade face.

Never slide a finger along the cutting edge.

To check with a fingertip, hold the knife still and move the fingertip very lightly from the blade face toward the edge, perpendicular to it. Stop as soon as you reach the apex. Repeat at the heel, middle, curved front and tip.

If you do not want to place a fingertip near the apex, use a small piece of cotton, tissue or thin paper. A burr may catch loose fibres more readily from one side. Good light or magnification can also help.

Do not confuse the burr with:

- grey slurry or steel residue stuck to the blade;
- rough scratches that feel the same from both directions;
- a rolled section that was already present before sharpening;
- a burr found at only one point, which proves only that one point reached the apex.

A large burr is not better. It can fold from side to side, cut paper briefly and then collapse in the first real use. Treat the burr as a receipt, not a trophy: the smallest continuous one is enough.

1.9 Three checks after sharpening

No single test proves that the whole knife is ready. Use all three.

1.9.1 Check 1: reflected light

Hold the clean, dry apex under a strong lamp and tilt it slowly. A clean apex appears mostly dark. A bright point or short bright line often marks a flat, rolled or chipped area.

One tiny reflection does not justify grinding the whole knife again. Note its position and compare it with the paper test.

1.9.2 Check 2: paper in three sections

Use ordinary dry printer paper. Hold it so your supporting hand remains well away from the cutting path.

Test three overlapping sections separately:

1. heel to middle;
2. middle to curved front;
3. curved front to tip.

Cut slowly instead of slashing. Notice how much force starts the cut, whether resistance changes and whether the paper is sliced cleanly or torn. The three parts should behave similarly. A catch in the same position twice identifies an area that needs attention.

Paper alone is not enough. A wire burr can cut it once and then fold.

1.9.3 Check 3: food and stability

A tomato checks whether the apex starts on smooth skin without sliding. An onion checks entry and movement through layers. A carrot also reveals thickness behind the microbevel, but that subject belongs to the next chapter.

Use the knife normally. Do not force it into food to prove that it is sharp.

After about twenty normal cuts through onion or carrot, clean the knife and repeat the three-part paper test. If the result becomes much worse immediately, a wire burr probably remained. Return only to the nearly weightless alternating passes, clean the knife and test again.

1.10 Workshop examples

1.10.1 The middle cuts, but the tip catches paper

The user held a useful angle on the straight part but did not begin with the handle raised enough to follow the tip.

Recolour the tip microbevel. Make three complete-edge passes per side. Start with the handle slightly raised, then lower it gradually through the curve. Test the tip again.

The beginner still covers the whole edge. Only the attention given to the tip changes.

1.10.2 The knife cuts paper once, then feels dull

A flexible wire burr probably remained. More heavy sharpening would make it larger.

Return to the medium stone and use six to ten nearly weightless alternating complete-edge passes. Clean the knife, make several normal food cuts and test again.

1.10.3 The marker disappears above the microbevel but remains at the edge

The angle is too low. The stone is polishing above the microbevel without reaching the apex.

Raise the spine slightly, recolour the last one or two millimetres and repeat one light contact pass.

1.10.4 The marker disappears only from the very edge

The spine is higher than the existing microbevel angle. This can create a steeper new microbevel instead of following the one already present.

Lower the spine slightly and repeat the marker check. Stop adjusting when the stone reaches the complete microbevel and apex.

1.10.5 The burr appears at the heel and middle but not at the tip

The tip has not reached the apex. Continue with two or three complete passes at a time. Begin with the handle slightly raised and watch the marker at the curved front.

Do not grind only the middle because its burr feels reassuring.

1.10.6 One side forms a burr quickly and the other takes longer

This is common on a previously uneven microbevel. Do not force equal stroke counts. Complete the burr on each side and then use the same light alternating finish.

1.10.7 The blade sticks and the movement becomes jerky

The stone may be dry or clogged, or the pressure may be too high. Add water, rinse the surface if needed and reduce pressure. A slow smooth pass sharpens better than a forceful pass that changes angle.

1.10.8 The knife cuts paper but wedges in a carrot

The microbevel may be sharp while the blade behind it is too thick. Repeated sharpening of the microbevel will not solve that resistance. This is a geometry problem and introduces thinning in the next chapter.

1.11 Mistakes that matter most

1.11.1 Sharpening the primary bevel instead of the microbevel

The primary bevel is the broad surface above the narrow terminal facet. Grinding it changes blade geometry and can damage a polished, hammered or Damascus finish. The beginner method touches only the microbevel.

1.11.2 Changing the angle during every pass

An approximate angle repeated well works better than a perfect angle touched only occasionally. Slow down and repeat the marker check.

1.11.3 Pressing harder when nothing happens

First check microbevel contact, water and stone condition. Pressure cannot compensate for missing the apex.

1.11.4 Stopping the pass in the middle

The complete-edge method works because every normal pass includes tip, middle and heel. Remove pressure, reset and repeat the full path.

1.11.5 Chasing a large burr

A large burr feels obvious but is harder to remove. Stop when the first light burr is continuous.

1.11.6 Testing only the middle

The middle is usually easiest to sharpen. Test the heel, curved front and tip separately as well.

1.11.7 Starting on a damaged or expensive knife

Large chips, unusual geometry and valuable finishes create problems that a first session does not need. Learn the movement on a simple knife first.

1.12 Base-level exercises

These exercises turn a lucky result into a repeatable skill.

1.12.1 Exercise 1: stable setup

Prepare the stone, then push gently on each corner and make ten unloaded practice movements. Pass when the stone stays still, the shoulders remain relaxed and the movement finishes without hitting an obstacle.

1.12.2 Exercise 2: the marker line

Colour only the microbevel. Make three very light complete-edge passes per side. Pass when ink is removed from the microbevel up to the apex at the heel, middle and tip without creating a wide scratched strip across the primary bevel.

If the result varies, recolour and repeat more slowly. Do not continue to burr formation until contact is understandable.

1.12.3 Exercise 3: the burr map

Sharpen one side in small groups and check heel, middle, curved front and tip after every group. Pass when you can identify the first light continuous burr and stop without making it much larger.

1.12.4 Exercise 4: pressure reduction

Say or note the three stages while working: working, refining and finishing. Pass when the final alternating movements are clearly lighter than the first passes and the knife no longer carries a directional burr.

1.12.5 Exercise 5: the three-part paper test

Test heel-middle, middle-front and front-tip. Pass when all three sections start and travel with similar resistance and none repeatedly tears the paper.

1.12.6 Exercise 6: edge stability

Make about twenty normal cuts in onion or carrot, clean the knife and repeat the paper test. Pass when the edge behaves substantially as it did before the food test.

1.12.7 Exercise 7: repeat the result

Sharpen a second ordinary knife, or repeat the process after the first knife has genuinely become dull again. Pass when the result is similar

without having to rediscover the movement from the beginning.

1.13 Your base-level test

Move to the next chapter only after you can answer yes to every statement in two different sharpening sessions:

- I can identify a suitable double-bevel practice knife and recognise the cases that require me to stop.
- I can prepare a flat, wet and stable stone without it moving.
- I can distinguish the primary bevel, microbevel and apex.
- I can use marker to place only the microbevel on the stone from tip to heel.
- Every normal working pass covers the complete edge.
- I begin with the handle slightly raised at the tip and lower it through the curve.
- I can form the smallest useful continuous burr and stop.
- I can reduce pressure and remove the directional burr with alternating passes.
- Heel, middle and tip pass the paper test with similar resistance.
- The result remains stable after normal food cutting.

Passing once can be luck. Passing in two sessions shows that the method is repeatable.

At this point you have learned the essential skill: controlled contact on the microbevel along the complete edge. The next chapter teaches when and how to depart from this safe beginner method.

Chapter 2

Improving Your Sharpening Skills

The first chapter gives one dependable answer to most beginner problems: follow the existing microbevel, cover the complete edge, form a light burr on both sides, then finish with very light alternating passes.

Keep that method whenever it gives a good result. Improvement does not mean replacing a simple method with a complicated one. It means noticing more, correcting less and understanding when the simple method needs help.

Begin this chapter only after you can sharpen an ordinary double-bevel knife evenly from heel to tip and repeat the result in two sessions. If microbevel contact, burr detection, the three-part paper test or the stability test is still uncertain, more advanced choices will hide the basic problem rather than solve it.

2.1 The three skills behind better sharpening

An improving sharpener learns to control three things:

1. **Contact:** exactly where the stone touches the knife.

2. **Pressure:** how strongly it cuts and how pressure changes during the session.
3. **Purpose:** whether the work is restoring the apex, correcting the blade behind it or changing the final cutting feel.

These skills reduce unnecessary grinding. They also explain why a knife can have a keen edge and still perform poorly in food.

2.2 Read the evidence after every stage

The stone leaves visible and tactile evidence:

- marker removal shows contact on the microbevel;
- fresh scratches show where steel was removed;
- reflected light reveals remaining flat or damaged areas;
- the burr confirms that the abrasive reached the apex;
- resistance in paper maps uneven sections;
- resistance in food reveals both the edge and the blade geometry behind it.

Do not collect all this information at the end. Stop, rinse and inspect during the work. Early inspection turns a correction into a few controlled passes. Late inspection often discovers that too much steel has already been removed.

2.2.1 Practice drill: predict before looking

Colour only the microbevel of a practice knife. Make one light pass while deliberately holding the spine slightly too low. Before inspecting it, predict where the ink should remain. Then repeat with the spine slightly too high.

This safe marker exercise teaches the connection between hand position and contact. Return to the correct angle and erase the intended strip evenly before sharpening further.

Pass the drill when your prediction matches the mark three times in a row.

2.3 Build three pressure levels

Pressure has different jobs during one sharpening session.

2.3.1 Working pressure

Use controlled pressure to remove steel and reach the apex. The stone should cut steadily without stopping the blade, moving the holder or making the wrist collapse.

2.3.2 Refining pressure

Once the burr is continuous, use about half the previous pressure. The goal is now to replace coarse scratches and reduce the burr, not to keep bending it.

2.3.3 Finishing pressure

The final alternating passes use little more than the weight of the knife and guiding fingers. They clean and centre the apex.

A beginner often applies one pressure from start to finish. The result may feel sharp while still carrying a wire burr. The clearest sign of progress is not faster movement; it is a visible reduction in pressure as the edge nears completion.

2.3.4 Practice drill: make the burr smaller

Sharpen two similar inexpensive knives on the same medium stone. On the first, stop at the earliest continuous burr. On the second, continue for twenty unnecessary firm passes.

Deburr both with the same light sequence. Compare how long the process takes and whether the second burr keeps folding from side to side. The extra work usually makes the finish harder rather than better.

The lesson is practical: the smallest burr that confirms complete contact is enough.

2.4 Introduce section work carefully

The complete-edge pass remains the safest default. Section work becomes useful when one area needs more attention, the blade is longer than the stone, or a local defect cannot be corrected efficiently with full passes.

Divide the edge into overlapping zones: heel, middle and front. Work one zone with short controlled strokes while keeping pressure directly above it. Overlap the neighbouring zone so no abrupt step develops. Finish with several light complete-edge passes to blend the whole edge.

Section work saves steel only when the sharpener can preserve the profile. Repeated strokes in one exact spot can create a hollow in the cutting line. Mark the microbevel, inspect often and stop local work as soon as the missing contact or burr appears.

2.4.1 Workshop example: one dull centimetre near the heel

The complete-edge beginner method would sharpen the whole knife until the heel improves. That is safe and may be the correct choice for a new sharpener.

An improving sharpener can make three to five short strokes over the heel, with the supporting fingers above it, then return to light full-edge passes. The correction ends when the burr and paper response match the rest of the blade.

The advanced method is not automatically better. It is better only when the user can keep the correction local without changing the profile.

2.4.2 Section-work test

Colour the full microbevel and leave one small area with extra marker. Work that area briefly, blend with full passes and inspect the edge line from the side.

Pass when the target mark is removed, the neighbouring microbevel remains even and the paper test has no catch at either transition.

2.5 Choose stones by the work

Grit numbers describe abrasive size, but they do not tell the whole story. Stone hardness, binder, abrasive type, cutting speed and slurry change how a stone behaves.

Use a simple three-stage decision:

- **Repair:** a coarse stone, usually below about 500 grit, for chips, profile correction or substantial thinning.
- **Sharpen:** a medium stone around 700-1500 grit for most dull edges.
- **Refine:** a finer stone around 2000-6000 grit when the task benefits from a smoother finish.

The ranges overlap between brands. Begin with the finest stone that can complete the required work in a reasonable time. A coarse stone saves time on damage but removes steel quickly. A fine stone gives control but cannot efficiently rebuild a badly rounded edge.

The abrasive must also cut the steel. Conventional water stones suit many common Xinzuo steels. Highly wear-resistant steels such as ZDP-189, SRS13, M390 and 14Cr14MoVNB may respond more efficiently to suitable ceramic or diamond abrasive. If the stone barely changes the microbevel despite correct contact, improve the abrasive before increasing force.

2.5.1 Hard and soft stones

A hard stone stays flat longer and shows contact clearly. This makes it useful for diagnosis and precise bevels. A softer stone releases abrasive readily and can feel smooth, but it dishes faster and can round a surface under heavy pressure.

Neither is a quality rank. Choose the behaviour that helps the stage you are performing.

2.5.2 Water, slurry and a readable surface

The grey mixture on a water stone contains removed steel and loose abrasive. It is called slurry or swarf depending on what it contains. It is not the burr.

Rinse the surface when you need to see exact contact, when the stone feels sticky or when metal begins to clog it. Keep a controlled slurry when the stone cuts and finishes better with it. A beginner may simply keep the surface wet and clean; deliberate slurry management belongs at this second level because the user can now notice the difference.

Clean the knife, hands and holder before moving to a fine stone. One coarse particle can leave a deep scratch across a polished microbevel or blade finish.

2.6 Keep the stone flat

The blade copies the surface beneath it. A hollow stone changes the angle during a stroke and makes accurate work harder.

Draw a pencil grid across the stone. Flatten until every line disappears. Check more often when using a soft stone, working a wide bevel or repeatedly sharpening the centre. Light, frequent flattening is faster than repairing a deeply dished stone.

2.6.1 Flatness test

Check the stone with pencil before a session, after sharpening two knives and again after flattening. Pass when you can recognise the developing wear pattern and correct it before the hollow becomes obvious under a straightedge.

2.7 Distinguish apex work from geometry work

Routine sharpening restores the narrow microbevel. It brings the two sides together at the apex.

As sharpening repeats, that microbevel moves upward into thicker steel and becomes wider. The knife may cut paper well but wedge in a carrot or onion. The apex is sharp; the blade behind it is too thick.

Thinning removes steel behind the edge to restore easier passage through food. It is a separate operation from sharpening and removes much more material.

2.7.1 A practical geometry comparison

Imagine two pencils sharpened to equally fine points. One tapers gradually; the other becomes thick immediately behind the point. Both points can pierce paper, but the gradual pencil passes through a narrow opening more easily.

A kitchen knife behaves in the same way. Paper mostly tests the point of the system. Firm food also tests the shape behind it.

2.7.2 Recognise when thinning may help

Possible signs include:

- the knife passes paper but splits or wedges in firm vegetables;
- the microbevel has become much wider after repeated sharpening;
- cutting resistance has increased even after a clean deburring;
- more force is needed once the apex has already entered the food.

Before thinning, inspect the blade finish. Damascus pattern, mirror polish, coatings, hammering and black forge scale can be scratched or removed. Protect valuable finishes and use professional service when the original grind is uncertain.

For a simple practice knife, colour the area immediately behind the microbevel. Lay the blade at a lower angle than the sharpening angle and

abrade that shoulder gradually. Keep the cutting profile unchanged. Then recreate the narrow microbevel and finish normally.

Do not try to make the visible microbevel equally wide at heel and tip. Blade thickness usually changes toward the tip. Equal appearance can require unequal geometry.

2.7.3 Geometry test

Compare the knife in printer paper and a firm carrot before and after normal sharpening. If paper improves but carrot resistance remains high, identify geometry as the likely next subject rather than moving automatically to a finer stone.

This test is passed when you can explain whether the problem begins at the apex or after the blade has entered the food. Actual thinning should wait until that distinction is clear.

2.8 Choose the finish for the food

Higher grit is not automatically sharper in every useful sense. A medium finish leaves microscopic texture that can start easily in tomato skin, meat and fibrous food. A finer finish can feel smoother in push cuts and delicate slicing.

Use the same knife and compare two finishes over time. Finish it first on the medium stone, deburr carefully and test tomato, onion and paper. At the next genuine sharpening, add the finer stone and repeat the same tests.

Keep the finish that suits the knife's work. Prestige, mirror appearance and grit number matter less than controlled cutting.

2.9 Improve deburring before adding polish

A wire edge can imitate excellent sharpness. It bends or breaks during the first cuts, so the knife seems to become dull immediately.

Prevent it by stopping at a small burr and reducing pressure early. Alternate sides more frequently near the end. Edge-leading strokes, where the edge moves into the abrasive, can cut weakened burr steel away efficiently. Edge-trailing strokes, where the spine leads, feel gentle but can draw out a new wire if pressure is high.

A clean leather or fabric strop is optional. Move spine first, use very few passes and press lightly. A soft strop under heavy force rounds the apex. The stone must do the main deburring; the strop is a final check, not a rescue tool.

2.9.1 Stability test

Cut paper, make twenty normal cuts through cardboard-free food such as onion or carrot, clean the knife and test paper again. Pass when the edge feels substantially the same and no section develops a sudden catch.

The test is not about long-term edge retention. It checks whether the first result depended on a weak burr.

2.10 Approach asymmetry with evidence

Some double-bevel knives have one side sharpened more than the other. Previous owners may also have created accidental asymmetry.

Do not equalise the two sides by counting identical strokes. Use marker, microbevel width, steering in food and burr movement. Preserve a deliberate asymmetry when the knife cuts straight and the maker specifies it. Correct an accidental one gradually rather than trying to rebuild it in a single session.

If the knife always steers to one side in a tall food such as potato, first confirm that the cut and grip are straight. Then compare microbevel angles, thickness and remaining burr. Steering is evidence, not a command to grind one side heavily.

2.11 Traditional single-bevel knives

A yanagiba, deba or usuba needs a different method. Its broad front bevel works with a hollow reverse called the **urasuki**. Narrow flat lands around that hollow support the edge; light work on them is called **uraoshi**.

Do not apply the beginner's double-bevel method to these knives. The reverse is not sharpened at a normal raised angle. It is placed almost flat on a truly flat stone and touched with very light pressure. Raising it can create a secondary bevel and damage the geometry.

The broad front bevel may also change contact from heel to tip because blade height, thickness and twist vary. A hard medium-fine stone and marker can reveal high and low areas before coarse work begins.

For an improving sharpener, the safe first lesson is diagnosis:

1. flatten the stone carefully;
2. place the reverse flat and make a few nearly weightless passes;
3. confirm that contact stays on the narrow perimeter lands rather than inside the hollow;
4. place the broad front bevel on the stone and inspect its contact pattern;
5. stop if the blade rocks, appears bent or would need extensive coarse correction.

Routine refinement of a sound single bevel can be learned with model-specific instruction. Correcting the reverse hollow, wide lands, serious twist or a damaged ridge line belongs to an experienced specialist.

2.11.1 Hamaguri in plain language

Hamaguri is a controlled convex transition on the front bevel. It can add support near an impact-prone edge, as on some deba sharpening strategies. It is not one compulsory shape for every Japanese knife.

Soft stones, thick slurry and rocking pressure can create accidental convexity. Learn flat contact first. A deliberate small microbevel is

easier to control than vaguely rounding the whole surface.

2.12 A five-session improvement plan

2.12.1 Session 1: contact

Use a practice knife and marker. Predict the ink pattern, make one light pass and inspect. Repeat until the hand can reproduce the same narrow contact strip.

2.12.2 Session 2: pressure

Sharpen normally, but name each stage aloud or in notes: working, refining, finishing. The final passes must be visibly lighter than the first.

2.12.3 Session 3: burr control

Check after every group of five passes. Stop at the first continuous burr and record how little work was actually needed. Test stability after deburring.

2.12.4 Session 4: local correction

Use marker to identify one incomplete section. Apply a few overlapping local strokes, then blend with complete-edge passes. Confirm the edge profile and paper response.

2.12.5 Session 5: task finish

Compare a medium and a fine finish on the same type of food. Describe bite, smoothness, resistance and control without using the grit number as the conclusion.

2.13 The second-level test

You have moved beyond beginner technique when you can demonstrate all of the following:

- predict and confirm contact with marker and scratches;
- use clearly different working, refining and finishing pressures;
- stop at a small continuous burr and produce a stable edge;
- correct one section without creating a hollow or transition catch;
- keep the stone flat and free from coarse contamination;
- distinguish a dull apex from excessive thickness behind the edge;
- choose a medium or fine finish by food performance;
- recognise a single-bevel knife and avoid damaging its reverse geometry.

Passing this test does not mean every repair should be attempted. It means you can read what the knife and stone are showing, choose a proportionate action and stop before uncertainty becomes damage.

2.14 What expert practice adds

Expert sharpening uses the same evidence with greater precision. The sharpener may change stroke direction to read scratches, select a hard stone to diagnose contact, use soft-stone slurry to blend a bevel, minimise burr formation or vary support along a twisted single bevel.

Those techniques are valuable because the foundation is stable. Without angle control, pressure control and reliable testing, extra choices only create extra ways to be inconsistent.

Keep the complete-edge method as your safe home position. Depart from it for a clear reason, make the smallest correction that evidence supports, then return to full light passes and test the whole knife.